

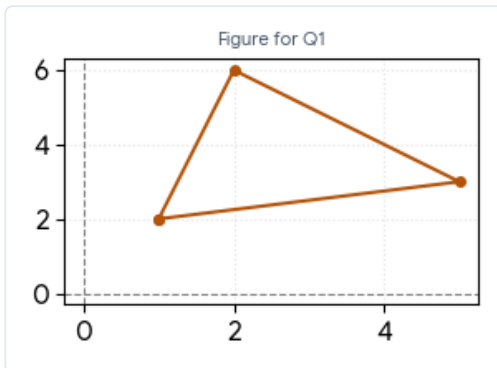
संभव BATCH - CRASH COURSE

Mathematics: Advanced Coordinate Geometry (with Figures)

Expert Guidance: Vipulendra Tripathi Sir

SSC CGL Tier-II Pattern Practice Set | 50 Hard Level Unique Questions | No Answer Key Included

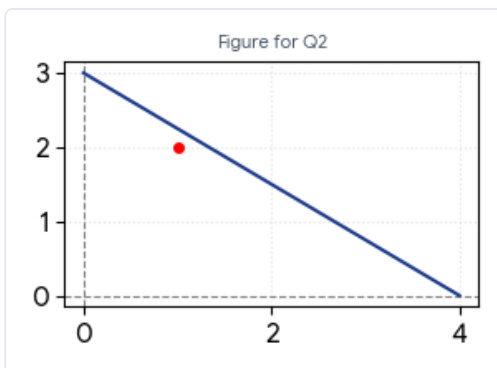
Q1. Find the coordinates of the orthocenter of a triangle whose vertices are plotted in the accompanying diagram as A(1, 2), B(2, 6), and C(5, 3).



- (A) (2, 3)
- (B) (3, 4)
- (C) $(\frac{7}{3}, \frac{11}{3})$
- (D) (2, 4)

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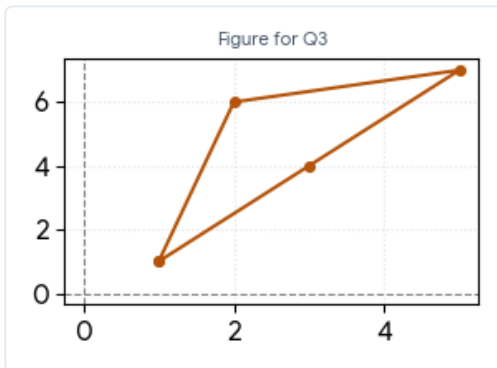
Q2. In the given coordinate layout, a line L_1 passes through (2, 3) and is perpendicular to $L_2: 3x - 4y + 5 = 0$. Find the reflection of the point (1, 2) in line L_1 .



- (A) (1, 1)
- (B) (2, 1)
- (C) $(\frac{7}{5}, \frac{6}{5})$
- (D) $(\frac{3}{5}, \frac{11}{5})$

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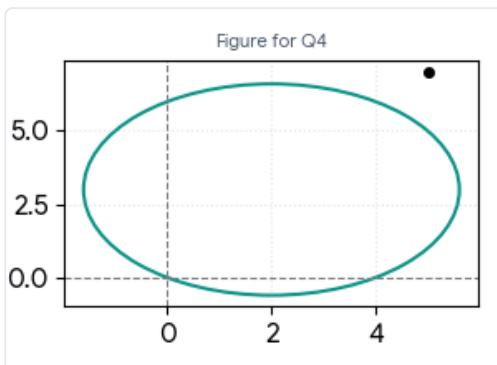
Q3. Find the area of the shaded quadrilateral shown below, whose coordinates are sequentially given as A(1, 1), B(3, 4), C(5, 7), and D(2, 6).



- (A) 6.5 sq units
- (B) 7.5 sq units
- (C) 4.5 sq units
- (D) 5.5 sq units

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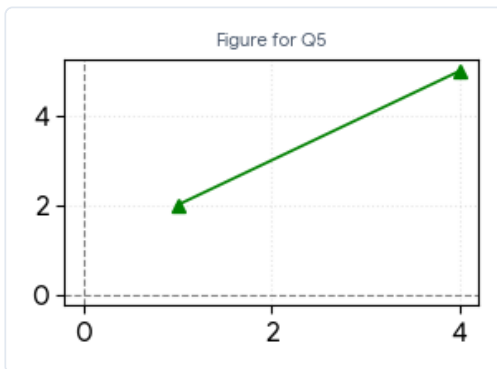
Q4. A circle passes through the points shown in the grid: (0, 0), (4, 0), and (0, 6). Find the length of the tangent drawn to this circle from the external point P(5, 7).



- (A) 5 units
- (B) $\sqrt{19}$ units
- (C) $\sqrt{17}$ units
- (D) $\sqrt{23}$ units

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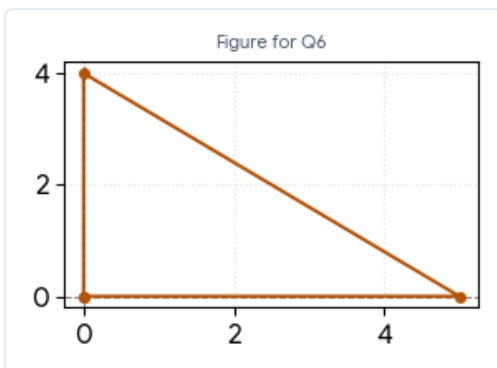
Q5. The line segment joining A(-2, 3) and B(4, 7) is divided internally by the y-axis in the ratio $m:n$, and externally by the x-axis in the ratio $p:q$ as demonstrated. Find the value of $(m/n) + (p/q)$.



- (A) $2/3$
- (B) $1/6$
- (C) $-1/3$
- (D) $5/6$

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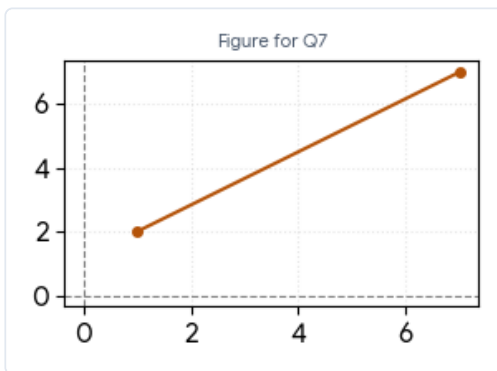
Q6. Calculate the incenter of the right-angled triangle plotted below with vertices (0,0), (5, 0), and (0, 4).



- (A) (2, 2)
- (B) (1.2, 1.2)
- (C) (1.5, 1.5)
- (D) (1, 1)

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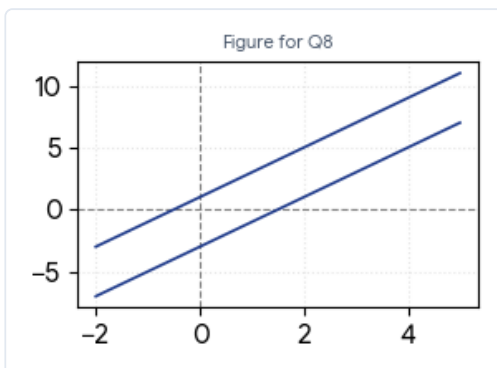
Q7. Find the ratio in which the line joining (1, 2) and (7, 7) is cut by the line $x + y = 6$.



- (A) 3:4
- (B) 5:7
- (C) 1:3
- (D) 2:5

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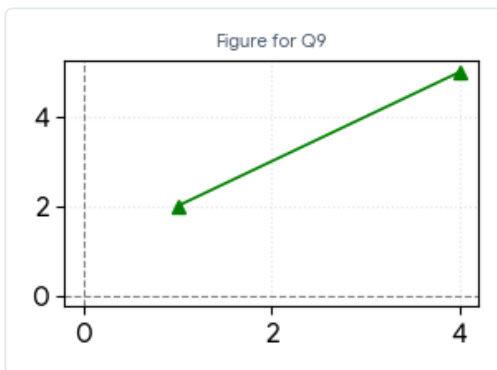
Q8. What is the distance between the parallel lines L1 and L2 shown in the figure?



- (A) $5/\sqrt{13}$ units
- (B) $2/\sqrt{5}$ units
- (C) $3/\sqrt{10}$ units
- (D) $4/5$ units

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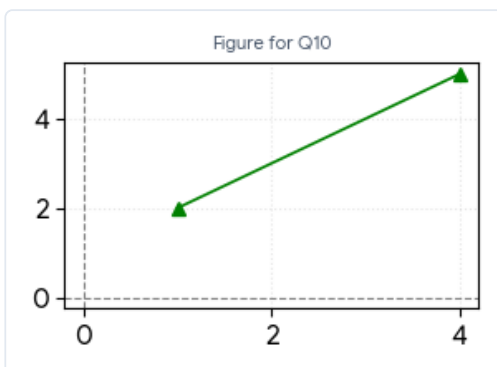
Q9. Find the area of the triangle formed by the straight line intercepts shown: $2x + 3y = 15$ with the coordinate axes.



- (A) 12 sq units
- (B) 7.50 sq units
- (C) 18.75 sq units
- (D) 18.5 sq units

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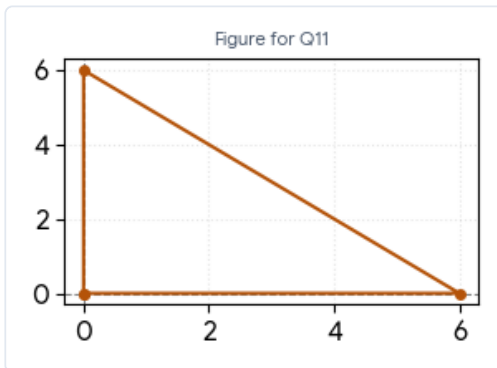
Q10. Find the equation of the locus of a point whose distance from the line $x - y = 0$ is always equal to its distance from the point plotted at $(2, 2)$.



- (A) $xy = 12$
- (B) $x^2 + y^2 + 2xy - 40x = 0$
- (C) $x^2 - y^2 = 4$
- (D) $x^2 + y^2 - 2xy - 4x - 4y = 0$

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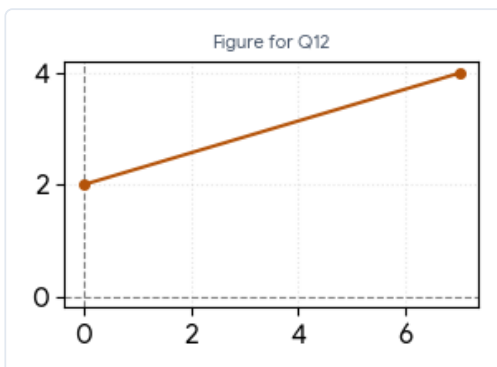
Q11. Calculate the incenter of the right-angled triangle plotted below with vertices $(0,0)$, $(6,0)$, and $(0,6)$.



- (A) $(1.2, 1.2)$
- (B) $(1, 1)$
- (C) $(2, 2)$
- (D) $(1.5, 1.5)$

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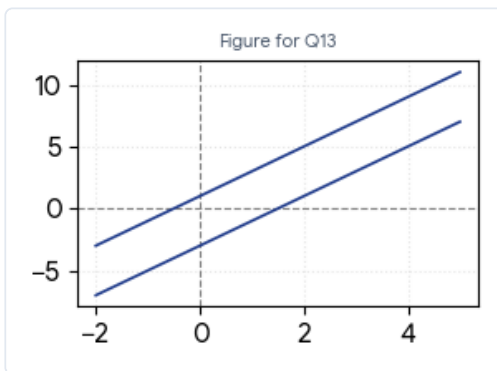
Q12. Find the ratio in which the line joining $(0, 2)$ and $(7, 4)$ is cut by the line $x + y = 6$.



- (A) 2:5
- (B) 5:7
- (C) 3:4
- (D) 1:3

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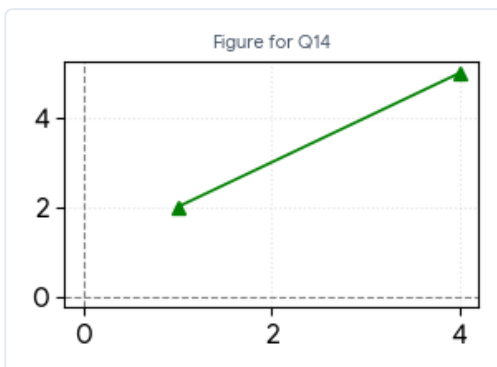
Q13. What is the distance between the parallel lines L1 and L2 shown in the figure?



- (A) $2/\sqrt{5}$ units
- (B) $5/\sqrt{13}$ units
- (C) $4/5$ units
- (D) $3/\sqrt{10}$ units

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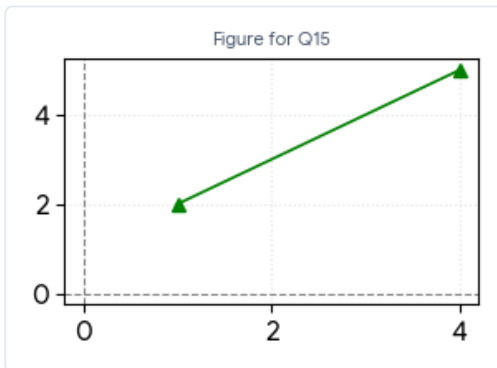
Q14. Find the area of the triangle formed by the straight line intercepts shown: $2x + 3y = 20$ with the coordinate axes.



- (A) 12 sq units
- (B) 10.00 sq units
- (C) 33.33 sq units
- (D) 18.5 sq units

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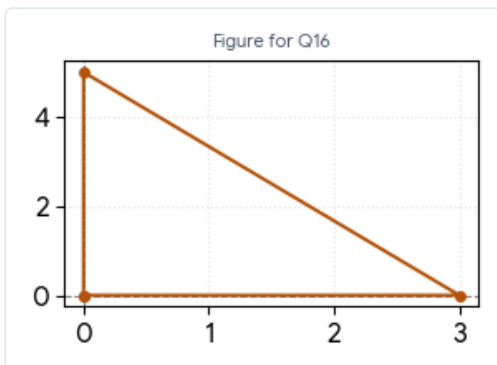
Q15. Find the equation of the locus of a point whose distance from the line $x - y = 0$ is always equal to its distance from the point plotted at $(1, 3)$.



- (A) $x^2 - y^2 = 4$
- (B) $x^2 + y^2 + 2xy - 60x = 0$
- (C) $xy = 12$
- (D) $x^2 + y^2 - 2xy - 4x - 4y = 0$

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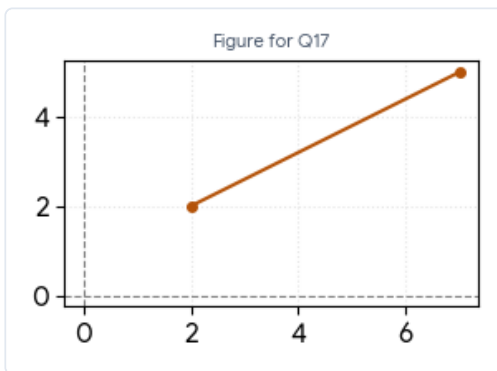
Q16. Calculate the incenter of the right-angled triangle plotted below with vertices $(0,0)$, $(3, 0)$, and $(0, 5)$.



- (A) $(2, 2)$
- (B) $(1.2, 1.2)$
- (C) $(1.5, 1.5)$
- (D) $(1, 1)$

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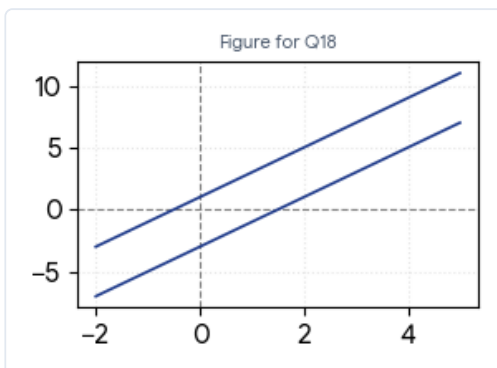
Q17. Find the ratio in which the line joining (2, 2) and (7, 5) is cut by the line $x + y = 6$.



- (A) 5:7
- (B) 3:4
- (C) 2:5
- (D) 1:3

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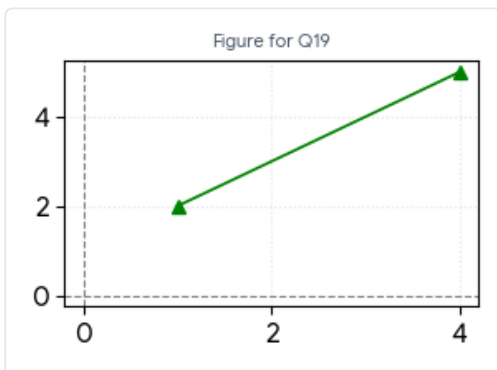
Q18. What is the distance between the parallel lines L1 and L2 shown in the figure?



- (A) $4/5$ units
- (B) $2/\sqrt{5}$ units
- (C) $5/\sqrt{13}$ units
- (D) $3/\sqrt{10}$ units

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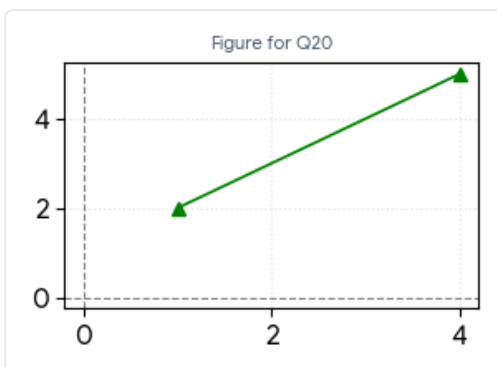
Q19. Find the area of the triangle formed by the straight line intercepts shown: $2x + 3y = 25$ with the coordinate axes.



- (A) 12.50 sq units
- (B) 52.08 sq units
- (C) 12 sq units
- (D) 18.5 sq units

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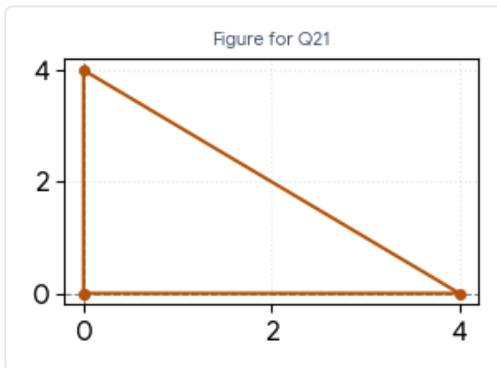
Q20. Find the equation of the locus of a point whose distance from the line $x - y = 0$ is always equal to its distance from the point plotted at $(3, 2)$.



- (A) $x^2 + y^2 - 2xy - 4x - 4y = 0$
- (B) $x^2 + y^2 + 2xy - 80x = 0$
- (C) $xy = 12$
- (D) $x^2 - y^2 = 4$

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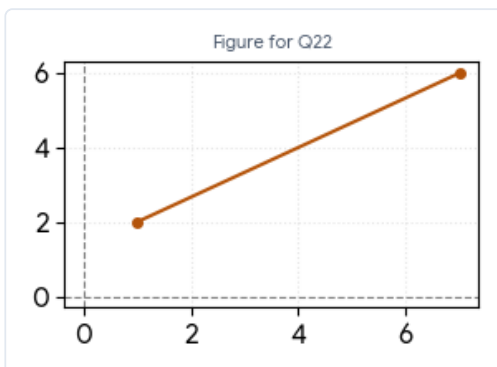
Q21. Calculate the incenter of the right-angled triangle plotted below with vertices $(0,0)$, $(4,0)$, and $(0,4)$.



- (A) $(2, 2)$
- (B) $(1, 1)$
- (C) $(1.2, 1.2)$
- (D) $(1.5, 1.5)$

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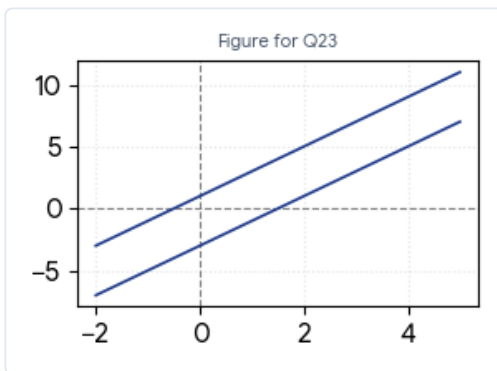
Q22. Find the ratio in which the line joining $(1, 2)$ and $(7, 6)$ is cut by the line $x + y = 6$.



- (A) 2:5
- (B) 5:7
- (C) 3:4
- (D) 1:3

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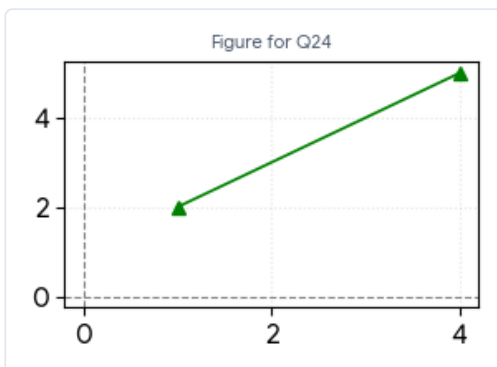
Q23. What is the distance between the parallel lines L1 and L2 shown in the figure?



- (A) $4/5$ units
- (B) $3/\sqrt{10}$ units
- (C) $2/\sqrt{5}$ units
- (D) $5/\sqrt{13}$ units

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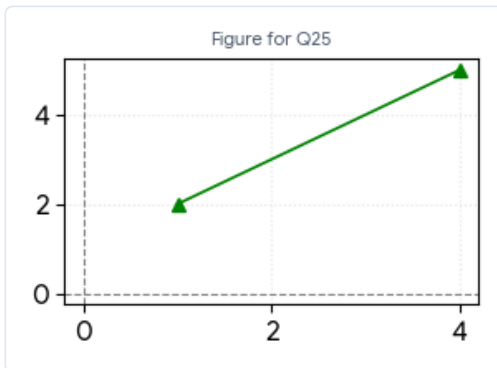
Q24. Find the area of the triangle formed by the straight line intercepts shown: $2x + 3y = 30$ with the coordinate axes.



- (A) 18.5 sq units
- (B) 75.00 sq units
- (C) 12 sq units
- (D) 15.00 sq units

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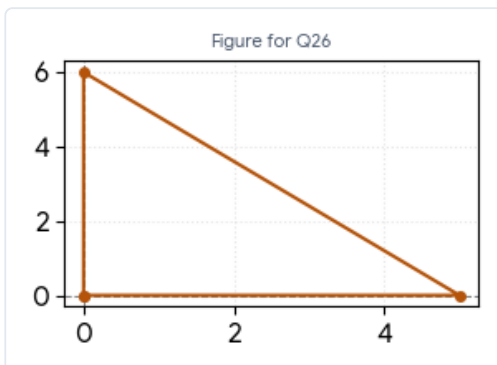
Q25. Find the equation of the locus of a point whose distance from the line $x - y = 0$ is always equal to its distance from the point plotted at $(2, 3)$.



- (A) $x^2 + y^2 + 2xy - 100x = 0$
- (B) $xy = 12$
- (C) $x^2 - y^2 = 4$
- (D) $x^2 + y^2 - 2xy - 4x - 4y = 0$

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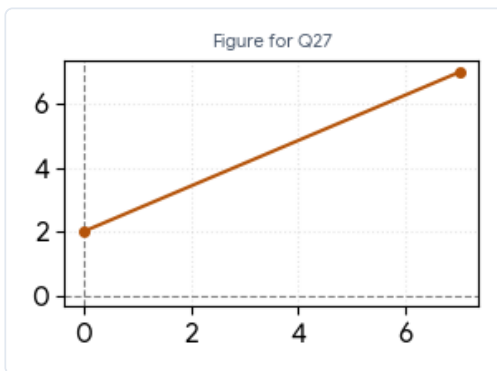
Q26. Calculate the incenter of the right-angled triangle plotted below with vertices $(0,0)$, $(5, 0)$, and $(0, 6)$.



- (A) $(2, 2)$
- (B) $(1, 1)$
- (C) $(1.5, 1.5)$
- (D) $(1.2, 1.2)$

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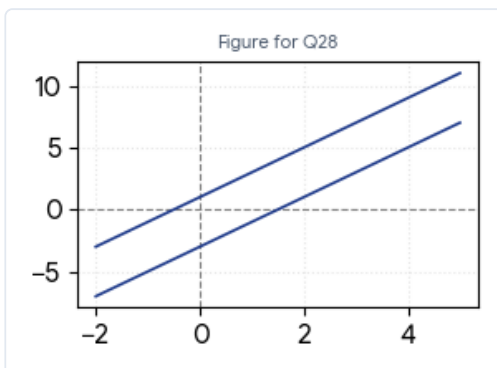
Q27. Find the ratio in which the line joining $(0, 2)$ and $(7, 7)$ is cut by the line $x + y = 6$.



- (A) 5:7
- (B) 3:4
- (C) 2:5
- (D) 1:3

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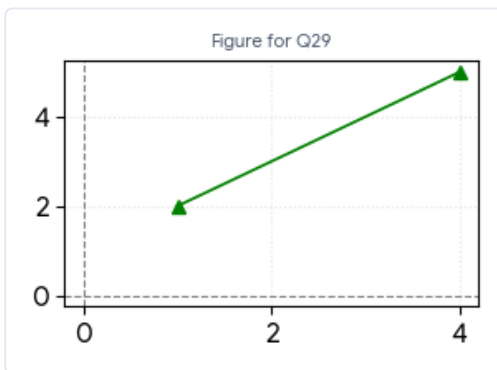
Q28. What is the distance between the parallel lines L_1 and L_2 shown in the figure?



- (A) $4/5$ units
- (B) $5/\sqrt{13}$ units
- (C) $3/\sqrt{10}$ units
- (D) $2/\sqrt{5}$ units

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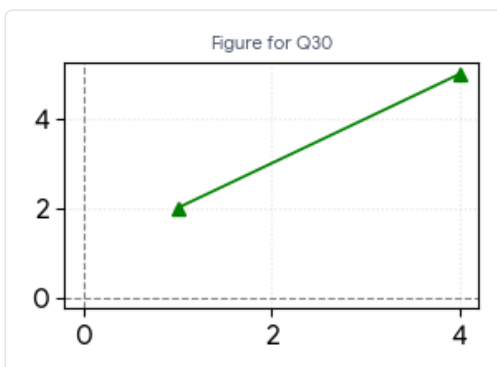
Q29. Find the area of the triangle formed by the straight line intercepts shown: $2x + 3y = 35$ with the coordinate axes.



- (A) 12 sq units
- (B) 102.08 sq units
- (C) 17.50 sq units
- (D) 18.5 sq units

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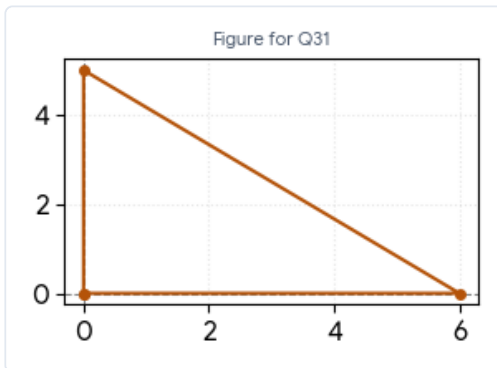
Q30. Find the equation of the locus of a point whose distance from the line $x - y = 0$ is always equal to its distance from the point plotted at $(1, 2)$.



- (A) $x^2 + y^2 + 2xy - 120x = 0$
- (B) $x^2 + y^2 - 2xy - 4x - 4y = 0$
- (C) $xy = 12$
- (D) $x^2 - y^2 = 4$

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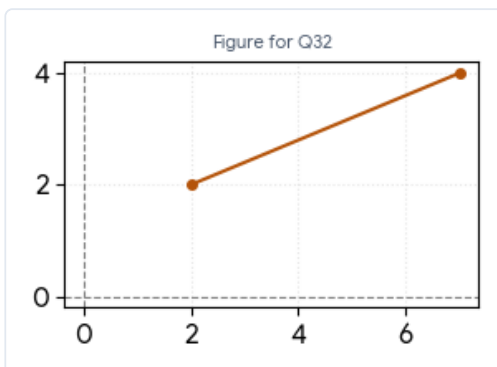
Q31. Calculate the incenter of the right-angled triangle plotted below with vertices $(0,0)$, $(6, 0)$, and $(0, 5)$.



- (A) $(1.5, 1.5)$
- (B) $(1, 1)$
- (C) $(1.2, 1.2)$
- (D) $(2, 2)$

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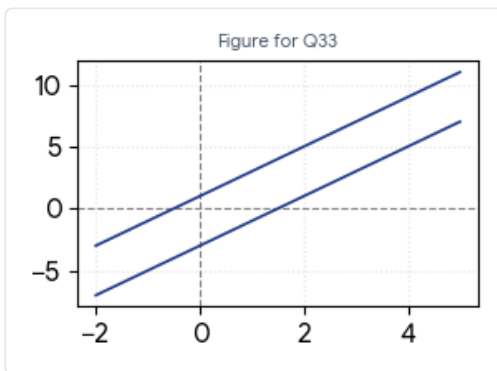
Q32. Find the ratio in which the line joining $(2, 2)$ and $(7, 4)$ is cut by the line $x + y = 6$.



- (A) 5:7
- (B) 3:4
- (C) 1:3
- (D) 2:5

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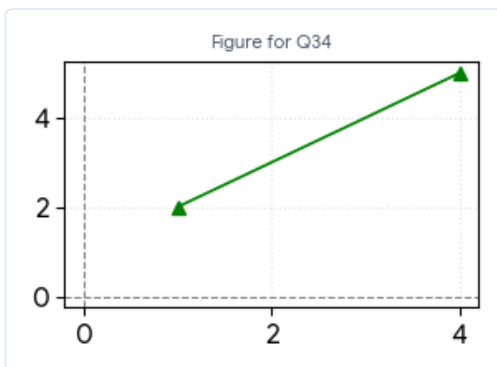
Q33. What is the distance between the parallel lines L1 and L2 shown in the figure?



- (A) $3/\sqrt{10}$ units
- (B) $2/\sqrt{5}$ units
- (C) $5/\sqrt{13}$ units
- (D) $4/5$ units

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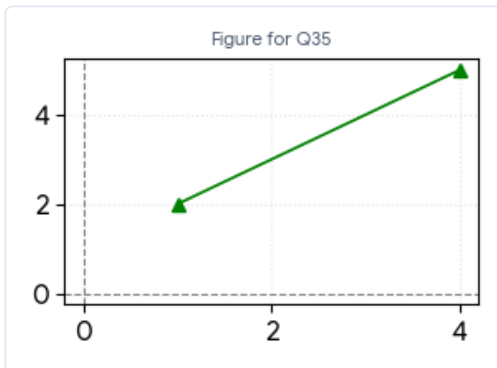
Q34. Find the area of the triangle formed by the straight line intercepts shown: $2x + 3y = 40$ with the coordinate axes.



- (A) 18.5 sq units
- (B) 12 sq units
- (C) 20.00 sq units
- (D) 133.33 sq units

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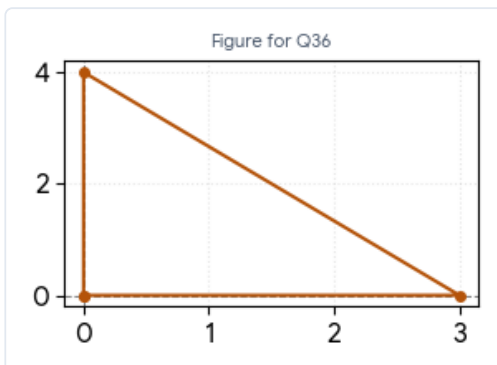
Q35. Find the equation of the locus of a point whose distance from the line $x - y = 0$ is always equal to its distance from the point plotted at $(3, 3)$.



- (A) $x^2 - y^2 = 4$
- (B) $xy = 12$
- (C) $x^2 + y^2 - 2xy - 4x - 4y = 0$
- (D) $x^2 + y^2 + 2xy - 140x = 0$

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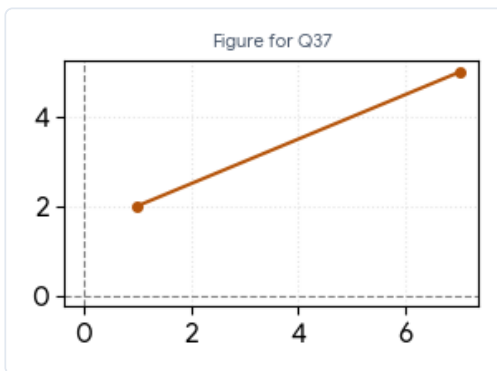
Q36. Calculate the incenter of the right-angled triangle plotted below with vertices $(0,0)$, $(3, 0)$, and $(0, 4)$.



- (A) $(2, 2)$
- (B) $(1, 1)$
- (C) $(1.2, 1.2)$
- (D) $(1.5, 1.5)$

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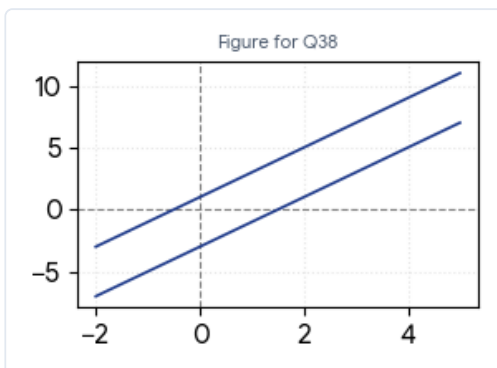
Q37. Find the ratio in which the line joining (1, 2) and (7, 5) is cut by the line $x + y = 6$.



- (A) 2:5
- (B) 5:7
- (C) 3:4
- (D) 1:3

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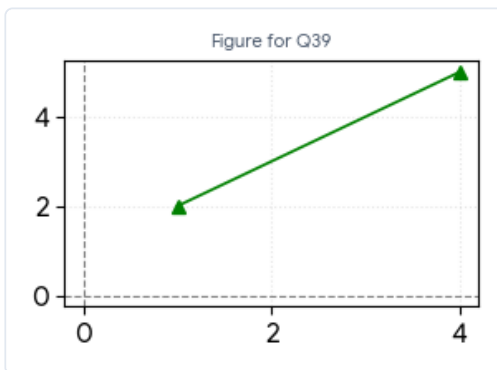
Q38. What is the distance between the parallel lines L1 and L2 shown in the figure?



- (A) $2/\sqrt{5}$ units
- (B) $5/\sqrt{13}$ units
- (C) $3/\sqrt{10}$ units
- (D) $4/5$ units

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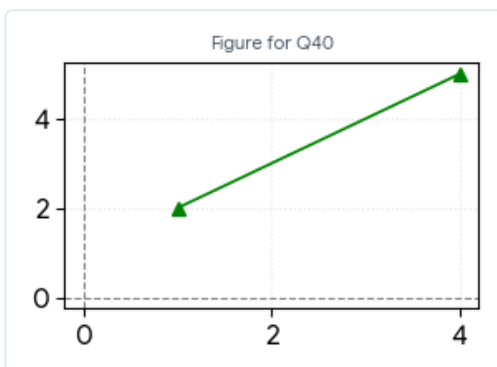
Q39. Find the area of the triangle formed by the straight line intercepts shown: $2x + 3y = 45$ with the coordinate axes.



- (A) 168.75 sq units
- (B) 12 sq units
- (C) 18.5 sq units
- (D) 22.50 sq units

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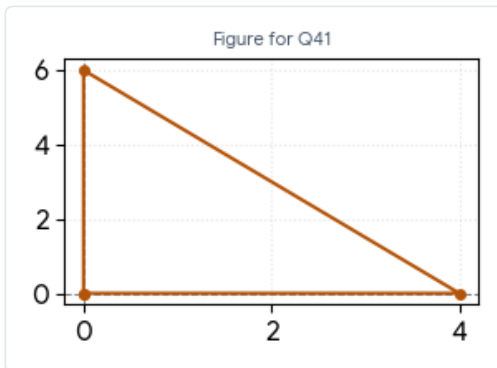
Q40. Find the equation of the locus of a point whose distance from the line $x - y = 0$ is always equal to its distance from the point plotted at $(2, 2)$.



- (A) $x^2 - y^2 = 4$
- (B) $xy = 12$
- (C) $x^2 + y^2 + 2xy - 160x = 0$
- (D) $x^2 + y^2 - 2xy - 4x - 4y = 0$

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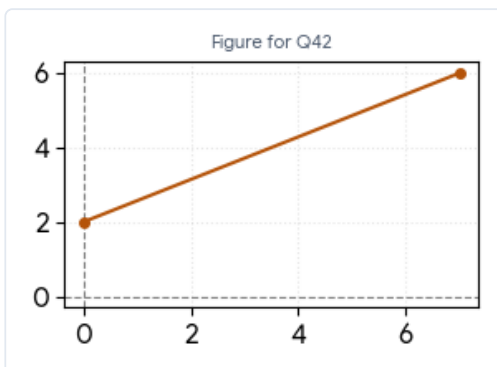
Q41. Calculate the incenter of the right-angled triangle plotted below with vertices $(0,0)$, $(4,0)$, and $(0,6)$.



- (A) $(1.5, 1.5)$
- (B) $(1.2, 1.2)$
- (C) $(2, 2)$
- (D) $(1, 1)$

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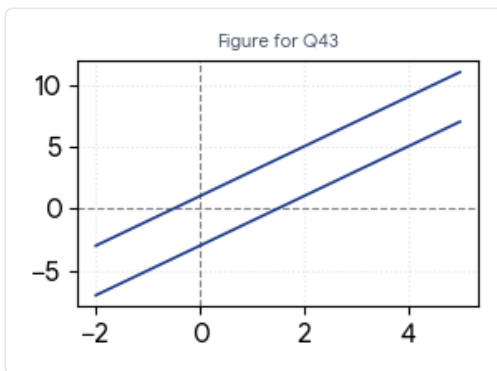
Q42. Find the ratio in which the line joining $(0, 2)$ and $(7, 6)$ is cut by the line $x + y = 6$.



- (A) 3:4
- (B) 5:7
- (C) 1:3
- (D) 2:5

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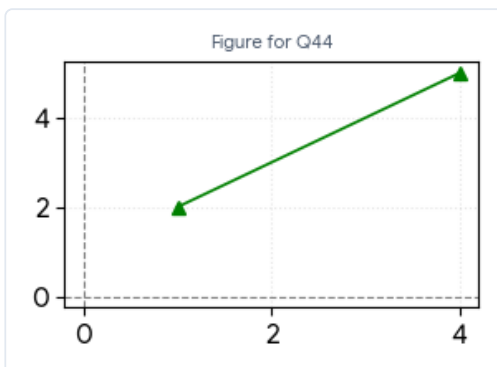
Q43. What is the distance between the parallel lines L1 and L2 shown in the figure?



- (A) $2/\sqrt{5}$ units
- (B) $4/5$ units
- (C) $5/\sqrt{13}$ units
- (D) $3/\sqrt{10}$ units

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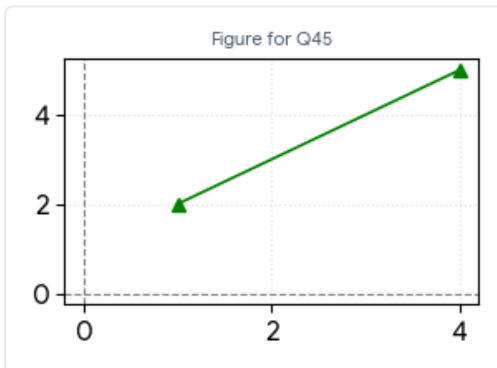
Q44. Find the area of the triangle formed by the straight line intercepts shown: $2x + 3y = 50$ with the coordinate axes.



- (A) 18.5 sq units
- (B) 12 sq units
- (C) 208.33 sq units
- (D) 25.00 sq units

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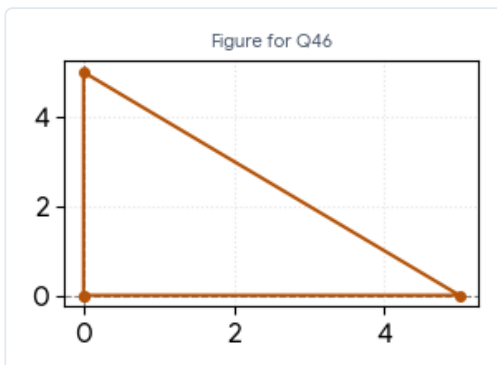
Q45. Find the equation of the locus of a point whose distance from the line $x - y = 0$ is always equal to its distance from the point plotted at $(1, 3)$.



- (A) $x^2 + y^2 - 2xy - 4x - 4y = 0$
- (B) $x^2 + y^2 + 2xy - 180x = 0$
- (C) $xy = 12$
- (D) $x^2 - y^2 = 4$

SSC CGL Tier-II 2021

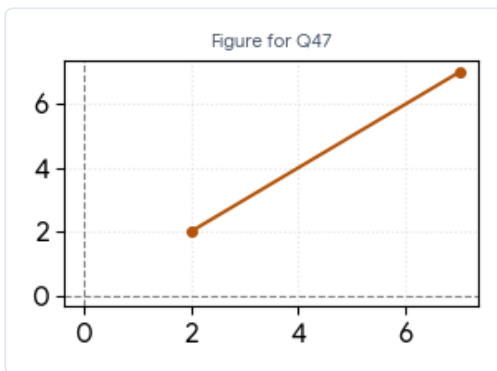
Q46. Calculate the incenter of the right-angled triangle plotted below with vertices $(0,0)$, $(5, 0)$, and $(0, 5)$.



- (A) $(1.5, 1.5)$
- (B) $(1, 1)$
- (C) $(2, 2)$
- (D) $(1.2, 1.2)$

SSC CPO 2022

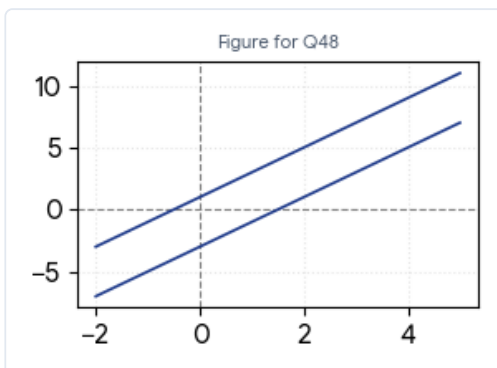
Q47. Find the ratio in which the line joining (2, 2) and (7, 7) is cut by the line $x + y = 6$.



- (A) 5:7
- (B) 1:3
- (C) 3:4
- (D) 2:5

SSC CHSL Tier-II 2022

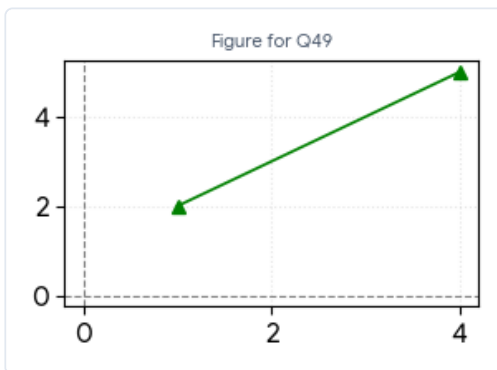
Q48. What is the distance between the parallel lines L1 and L2 shown in the figure?



- (A) $5/\sqrt{13}$ units
- (B) $3/\sqrt{10}$ units
- (C) $2/\sqrt{5}$ units
- (D) $4/5$ units

SSC Selection Post Phase XI

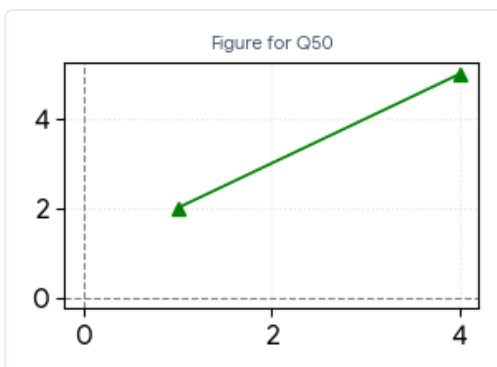
Q49. Find the area of the triangle formed by the straight line intercepts shown: $2x + 3y = 55$ with the coordinate axes.



- (A) 18.5 sq units
- (B) 252.08 sq units
- (C) 27.50 sq units
- (D) 12 sq units

SSC CGL 2022

Q50. Find the equation of the locus of a point whose distance from the line $x - y = 0$ is always equal to its distance from the point plotted at (3, 2).



- (A) $x^2 - y^2 = 4$
- (B) $xy = 12$
- (C) $x^2 + y^2 - 2xy - 4x - 4y = 0$
- (D) $x^2 + y^2 + 2xy - 200x = 0$

SSC CGL Tier-II 2021